Differential Equation: Formula Sheet

Order, degree, formation

Order: highest derivative present

Degree: power of highest derivative (polynomial form)

Formation: n constants $\to$ differentiate n times, order n

Variable separable

Separable: $\int f(x)\,dx = \int g(y)\,dy + c$

$y' = f(ax+by+c)$: $t = ax+by+c,\ t' = a+bf(t)$

$a_2 + b_1 = 0$: group $a_2\,d(xy)$

Polar 1:

$$x\,dx + y\,dy = r\,dr,\ x\,dy - y\,dx = r^2 d\theta$$

Polar 2:

$$x\,dx - y\,dy = r\,dr,\ x\,dy - y\,dx = r^2\sec\theta\,d\theta$$

Tangent and normal lengths

Tangent intercepts: $x - \frac{y}{m},\ \ y - mx$

Normal x-intercept: $x + my$

Subtangent $\left|\frac{y}{m}\right|$, subnormal $|ym|$

Lengths:

$$T = \left|\frac{y}{m}\right|\sqrt{1+m^2},\ N = |y|\sqrt{1+m^2}$$

Polar radius: $OP = \sqrt{x^2+y^2}$

Homogeneous

$y = vx$: $\int \frac{dv}{F(v)-v} = \ln|x| + c$

Shift: $x = X+h,\ y = Y+k$ ((h,k) = intersection)

Linear and reducible

Linear:

$$y\,e^{\int P\,dx} = \int Qe^{\int P\,dx}dx + C$$

Linear in x:

$$x\,e^{\int P_1 dy} = \int Q_1 e^{\int P_1 dy}dy + C$$

Bernoulli: $z = y^{1-n}$, $z' + (1-n)Pz = (1-n)Q$

Change of variable: $t = f(y)$ in $f'(y)y' + Pf(y) = Q$

Inequality: $f' + Pf > 0 \Rightarrow f\,e^{\int P}$ increasing

Rate problems

Growth/decay: $N = N_0 e^{kt}$

Cooling: $T - T_s = (T_0 - T_s)e^{-kt}$

Mixture:

$$Q' = \text{in} - \frac{Q}{V}\cdot\text{out}$$

Trajectories and exact forms

Orthogonal trajectory: $\frac{dy}{dx} \to -\frac{dx}{dy}$

$$d(xy) = x\,dy + y\,dx$$

$$d\left(\frac{y}{x}\right) = \frac{x\,dy - y\,dx}{x^2},\ d\left(\frac{x}{y}\right) = \frac{y\,dx - x\,dy}{y^2}$$

$$d\left(\tan^{-1}\frac{y}{x}\right) = \frac{x\,dy - y\,dx}{x^2+y^2}$$

$$d\left(\ln\frac{y}{x}\right) = \frac{x\,dy - y\,dx}{xy},\ d(\ln xy) = \frac{x\,dy + y\,dx}{xy}$$

$$d\left(\frac{1}{2}\ln(x^2+y^2)\right) = \frac{x\,dx + y\,dy}{x^2+y^2}$$

$$d\left(\frac{e^x}{y}\right) = \frac{ye^x\,dx - e^x\,dy}{y^2}$$